

Determinant

- Equations reducible to linear equation.
Ex:
Put to get linear equations as:
 $a + b = 5$; $2a - b = 7$

Applications

- Assume two variables x and y .
- From given information frame two equations in x and y .
- Solve the equations.

Linear Equation in Two Variables

General Form

$$ax + by + c = 0 \quad a, b, 0$$

Methods of Solving

Graphical

- Find three or four ordered pairs for given equations.
- Draw x -axis, y -axis on graph paper.
- See that all point lies on the same line.

Substitution

- Select any one equation out of two given equations.
- Find value of x/y and give it equation number (iii)
- Substitute eq(iii) in another equation which was not selected earlier
- You will get value of x/y
- Put this value of x/y in eq(iii) to get value of y/x

Elimination Method

- Write both equations in standard form.
- Make the coefficients of one variable opposite.
- Add the equations resulting from Step 2 to eliminate one variable.
- Solve for the remaining variable.
- Substitute the solution from Step 4 into one of the original equations.
- Write the solution as an ordered pair.

Determinant (Cramer's Rules)

- Write equations in the form $ax + by = c$
- Find the values of D , D_x and D_y
- Using $x = \frac{D_x}{D}$ and $y = \frac{D_y}{D}$ Find value of x and y .

Nature of Root

For $ax^2 + bx + c = 0$
 $D = b^2 - 4ac$

$D > 0$

Roots are real and distinct

$D = 0$

Roots are real and equal

$D < 0$

Roots are real not equal

- If α and β are roots of $ax^2 + bx + c = 0$ then $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$
- If α and β are given then quadratic equation can be formed as $ax^2 - (\alpha + \beta)x + \alpha\beta = 0$

Factorisation method

$ax^2 + bx + c = 0$
 (Splitting middle term)

$ax^2 + bx = 0$
 (Taking x common)

$ax^2 + c = 0$
 use: $a^2 - b^2 = (a+b)(a-b)$

Completing square method

- Steps:
 - Make sure coefficient of x^2 is 1.
 - Shift constant term to right side
 - Find third term: $(\frac{b}{2a})^2$
 - Add third term on both sides.
 - Complete square formula on left side
 - Take square root
 - Find value of x.

Formula method

- Compare given quadratic equation with General form.
- Find values of a, b, c
- Use formula to find roots: x

Quadratic Equation

Equation of degree two in one variable

General Form

$a \cdot x^2 + bx + c = 0$
 a, b, c Real number
 $a \neq 0$

Solution of Quadratic equation

Arithmetic mean

If a, b, c are three terms in an A.P.
then $b =$
 $b =$ arithmetic means of a and c

Definition

List of number in which each term
is obtained by adding a fixed
number to the previous term.

Find t_n if S_n is given

$$t_n = S_n - S_{n-1}$$

General Form

$$a, a + d, a + 2d, \dots, a + (n - 1)d$$

Selection of Terms in an A.P.

The following table shows
the convenient way of
choosing required number
of terms in A.P

Common difference (d)

The difference between
any two consecutive terms

Arithmetic Progression

Sum of first n terms

n^{th} term

When first term and common difference is given

$$S_n = [2a + (n-1)d]$$

a = first term
 d = common difference
 n = total terms

When first and last term is given

$$S_n = [2a + (n-1)d]$$

t_a = first term
 d = common difference
 n = total terms

From beginning

$$t_n = a + (n-1)d$$

a = first term
 d = common different
 t_n = n^{th} term

From the end

$$t_n = l - (n-1)d$$

l = last term
 d = common different
 t_n = n^{th} term

Probability

Random Experiment

The experiment in which all possible results are known but non can be predicted with certainty.

Outcome

Result of random experiment

Equally Likely Outcome

A given number of outcomes are said to be equally likely if one of them occurs.

Same Space

The set of all possible outcomes of random experiment.

Event

A set of all favourable outcome of given sample space

Probability of an event

$$P(A) = \frac{\text{No. of sample points in event A}}{\text{No. of sample points in sample space}} = \frac{n(A)}{n(S)}$$

